

32 **Abstract**

33 **Background.** The linear mixed model is the standard tool for testing a fixed-
 34 effect interaction in longitudinal and clustered data. The model-based Wald test
 35 of that interaction is exact only when the working within-cluster covariance is
 36 correctly specified. In practice an analyst chooses a convenient parametric residual
 37 structure, a continuous-time autoregressive correlation, compound symmetry, or
 38 an unstructured form, and then trusts the model-based standard error. The
 39 direction and the magnitude of the mis-calibration that follows when the working
 40 covariance is wrong are rarely checked.

41 **Methods.** We set out a staged diagnostic protocol, a four-channel decomposition
 42 of the test statistic, a working-covariance model ladder, and a sample-size gradient,
 43 and apply it to a worked example: the test of a biomarker-by-treatment interaction
 44 in an aggregated N-of-1 carryover simulation, analysed with an `lme` model carrying
 45 a random intercept and a `corCAR1` residual correlation.

46 **Results.** At the null the model-based standard error of the interaction is over-
 47 estimated by 6 to 10 percent, and the realised type-I error is approximately 0.03
 48 against a nominal 0.05. The degrees-of-freedom rule, the positive-definiteness
 49 correction, and point-estimate bias are excluded as causes. The model ladder
 50 localises the mis-calibration to the `corCAR1` residual layer: a random intercept
 51 alone is well calibrated. The sample-size gradient shows the bias is asymptotic,
 52 not a moderate-sample artefact: it does not attenuate from $N = 35$ to $N =$
 53 280 . A CR2 bias-reduced cluster-robust standard error restores the standard-
 54 error ratio to approximately one and the type-I error to approximately 0.05. At
 55 an alternative cell with a non-null effect the recalibration raises power by a near-
 56 uniform increment and leaves the ranking of analysis specifications unchanged.

57 **Conclusions.** A mis-specified parametric working covariance can bias the model-
 58 based standard error of a mixed-model interaction test asymptotically, in either
 59 direction. Cluster-robust standard errors are a general remedy, and the staged
 60 diagnostic is a reusable instrument for detecting the problem.

61 **1 Introduction**

62 The linear mixed model is, by a wide margin, the most common vehicle for testing
 63 whether an effect is moderated by a covariate in longitudinal or clustered data^{1,2}.
 64 The quantity of interest is a fixed-effect interaction coefficient, and the inference
 65 almost always proceeds through a Wald test: the coefficient estimate is divided by
 66 a model-based standard error and referred to a t distribution. That procedure is
 67 exact only under a condition that is easy to state and easy to forget. The working
 68 covariance, the random-effects structure together with the parametric residual
 69 correlation the analyst has assumed, must be correct.

70 It has been understood since the work of Eicker, Huber, and White that this
71 condition is more demanding than it appears³⁻⁵. When the working covariance is
72 mis-specified, the model-based standard error is no longer a consistent estimator of
73 the true sampling variability of the fixed-effect estimate. Liang and Zeger⁶ carried
74 this insight into longitudinal data analysis: their generalized estimating equations
75 decouple the point estimate, computed under a convenient working correlation,
76 from the variance estimate, computed by an empirical sandwich that remains
77 consistent whatever the working correlation. The sandwich, or cluster-robust,
78 variance estimator has since become a standard instrument, and a substantial
79 literature has refined it for the small-sample case⁷⁻¹⁰.

80 Yet in routine mixed-model practice the working covariance is still trusted. An
81 analyst fitting longitudinal data with `nlme` or a comparable package will choose a
82 residual correlation, very often a continuous-time first-order autoregressive form,
83 because it is pharmacokinetically or biologically plausible¹¹, and will then report
84 the model-based standard error of the interaction without asking what happens
85 if the chosen form is not exactly the form the data follow. This is not negligence
86 so much as a reasonable working assumption that is seldom audited. But how
87 large is the error when the assumption fails? Is the resulting test conservative or
88 anti-conservative? Does the mis-calibration diminish as the sample grows, so that
89 it can be dismissed as a small-sample nuisance, or does it persist? These questions
90 do not have a single answer, because the answer depends on the interaction being
91 tested and on the way the working covariance departs from the truth. They have
92 to be asked of a particular analysis.

93 This paper does exactly that for one analysis, and in doing so offers two contribu-
94 tions that we believe are more general than the single case. The first is a diagnostic
95 protocol. We decompose the Wald statistic into four channels through which mis-
96 calibration can enter, and we add two further instruments, a working-covariance
97 model ladder that localises the cause and a sample-size gradient that separates
98 a finite-sample artefact from an asymptotic one. The protocol is not specific to
99 the example and can be applied to any mixed-model interaction test whose cali-
100 bration is in doubt. The second contribution is the finding the protocol produced.
101 In our worked example the model-based interaction test is conservative, not anti-
102 conservative as the sandwich literature's emphasis on under-estimated standard
103 errors might lead one to expect; the conservatism is material, on the order of ten
104 percent in standard-error terms; and it is asymptotic, persisting undiminished
105 across an eightfold range of sample size. We trace it to a single component of the
106 working covariance, and we show that a cluster-robust standard error removes it.

107 The worked example is drawn from the `pmsimstats-ng` project, a simulation study
108 of power to detect biomarker-treatment interactions in aggregated N-of-1 trials
109 under carryover¹². The example is incidental to the argument. The phenomenon,
110 a parametric working covariance that biases a fixed-effect interaction test asymp-
111 totically, and the remedy are not properties of that simulation; they are properties
112 of the mixed-model Wald test, and the example serves only to make them con-
113 crete and quantified. We begin in Section 2 with the diagnostic protocol. Section

3 describes the worked example. Section 4 reports the staged diagnosis and the cluster-robust remedy. Section 5 discusses the generality of the result and its limits.

2 A staged diagnostic protocol

Consider a fixed-effect interaction coefficient θ estimated by $\hat{\theta}$, with a model-based standard error $\widehat{\text{SE}}$ and an assigned degrees of freedom ν . The Wald test rejects the null $H_0 : \theta = 0$ at level α when the statistic $T = \hat{\theta}/\widehat{\text{SE}}$ exceeds the critical value $t_{\nu, 1-\alpha/2}$ in absolute value. The test is correctly calibrated when the realised rejection rate under H_0 equals α . When it does not, the discrepancy must enter through one of exactly four channels, and the channels are separable by simulation.

2.1 The four-channel decomposition

We begin by writing the interaction estimate in a form that exposes the four channels. Let

$$\hat{\theta} = \mu + \sigma Z^*, \quad \mu = \text{E}(\hat{\theta}), \quad \sigma = \text{SD}(\hat{\theta}),$$

so that $Z^* = (\hat{\theta} - \mu)/\sigma$ is the standardised estimate, of mean zero and unit variance by construction; when $\hat{\theta}$ is approximately normal, Z^* is approximately standard normal. Write $\widehat{\text{SE}}$ for the mean of the model-based standard error and $r = \widehat{\text{SE}}/\sigma$ for its ratio to the true sampling standard deviation σ . Then the Wald statistic, to the order at which the variability of $\widehat{\text{SE}}$ about its mean may be neglected, is

$$T = \frac{\hat{\theta}}{\widehat{\text{SE}}} \approx \frac{\mu + \sigma Z^*}{r \sigma} = \frac{1}{r} (Z^* + \delta), \quad \delta = \frac{\mu}{\sigma},$$

where δ is the bias expressed in standard-deviation units. The realised type-I error follows at once,

$$\alpha_{\text{real}} = \Pr(|T| > t_{\nu, 1-\alpha/2}) \approx \Pr(|Z^* + \delta| > r t_{\nu, 1-\alpha/2}),$$

and exactly four quantities appear in it. They are the four channels: the standardised bias δ , which decentres the statistic; the scale r , which multiplies the rejection threshold; the degrees of freedom ν , which fix the critical value $t_{\nu, 1-\alpha/2}$; and the distribution of Z^* , the shape of the reference. When all four take their nominal values, $\delta = 0$, $r = 1$, $\nu = \infty$, and $Z^* \sim \text{N}(0, 1)$, the expression collapses to $\Pr(|\text{N}(0, 1)| > z_{1-\alpha/2}) = \alpha$ and the test is exact; a departure in any one channel mis-calibrates it. A second identity is worth recording, since the diagnosis turns on it: because $T \approx (Z^* + \delta)/r$, the dispersion of the statistic satisfies

$$\text{SD}(T) \approx \text{SD}(Z^*)/r,$$

so a statistic standard deviation below one is read at once as a scale inflation ($r > 1$), a sub-normal reference shape, or both.

We measure the four channels by simulation as follows.

- 145 1. **Point-estimate bias** (δ). If $E(\hat{\theta}) \neq 0$ under H_0 the statistic is decentred.
 146 We measure μ , and hence $\delta = \mu/\sigma$, by the mean of $\hat{\theta}$ across null replicates.
- 147 2. **Standard-error calibration** (r). If the model-based $\widehat{SE}$ does not estimate
 148 the true sampling standard deviation of $\hat{\theta}$, the statistic is rescaled. We
 149 measure r as the mean model-reported standard error over the empirical
 150 standard deviation of the estimate across replicates. A ratio above one
 151 indicates an over-stated standard error and a conservative test; a ratio below
 152 one the reverse.
- 153 3. **Degrees of freedom** (ν). If ν is too small the t reference has tails too
 154 heavy and the test is conservative; if too large, the reverse. We measure
 155 this by recomputing the p-values against a normal reference and comparing
 156 the realised type-I error.
- 157 4. **Reference-distribution shape** (the law of Z^*). Even with $\delta = 0$, $r =$
 158 1 , and ν correct, Z^* may not be standard normal. We measure this by
 159 the standard deviation of T under H_0 , which should be approximately one,
 160 and by the empirical distribution function of the p-values, which should be
 161 $\text{Uniform}(0, 1)$.

162 The individual channels are not new. The bias of the estimate and the ratio of the
 163 model-based to the empirical standard error are among the standard performance
 164 measures of a simulation study, as codified by Morris, White, and Crowther¹³,
 165 who recommend that the model-based and empirical standard errors be reported
 166 and compared as a matter of course; the degrees of freedom of the reference t
 167 are the subject of the small-sample literature of Satterthwaite¹⁴ and Kenward
 168 and Roger¹⁵; and the uniformity of the null p-value distribution is the elementary
 169 calibration check that follows from the probability integral transform. What the
 170 decomposition contributes is not a new measure but the observation that these
 171 four quantities, and exactly these four, compose into the single expression for α_{real}
 172 derived above. Two consequences follow. First, the channels are exhaustive: a
 173 Wald test is a centred statistic, a scale, a reference shape, and a degrees-of-freedom
 174 choice, and the realised error rate is a function of those and of nothing else, so a
 175 discrepancy absent from all four cannot exist. Second, because the channels enter
 176 the expression in distinct and non-interacting roles, measuring all four does not
 177 merely detect a mis-calibration but localises it, identifying which term carries the
 178 fault. Since the channels have different causes and different remedies, a diagnosis
 179 is in this sense a decomposition and not a checklist.

180 2.2 The working-covariance model ladder

181 The four-channel decomposition says *how* a test is mis-calibrated. It does not say
 182 *why*. When the decomposition points to the standard-error channel, it is worth
 183 being precise about what the scale r measures. Let the interaction estimate be
 184 written $\hat{\theta} = \ell' \hat{\beta}$, with ℓ the contrast vector that selects the interaction coefficient
 185 from the fixed-effects vector. A fixed-effect estimator that solves a generalized
 186 least squares system under a working covariance V_0 , when the data are in truth

187 generated under a covariance V , has a model-based variance

$$v_0 = \ell'(X'V_0^{-1}X)^{-1}\ell,$$

188 while its true sampling variance is the sandwich form

$$v = \ell'(X'V_0^{-1}X)^{-1}(X'V_0^{-1}VV_0^{-1}X)(X'V_0^{-1}X)^{-1}\ell,$$

189 where X is the fixed-effects design and, the clusters being independent, every
190 cross-product is a sum over clusters, $X'V_0^{-1}X = \sum_i X_i'V_{0i}^{-1}X_i$ and likewise
191 $X'V_0^{-1}VV_0^{-1}X = \sum_i X_i'V_{0i}^{-1}V_iV_{0i}^{-1}X_i$ ^{5,6}. The scale of Section 2.1 is the ratio
192 of these two quadratic forms, $r^2 = v_0/v$. They coincide when $V_0 = V$, and for
193 a generic contrast ℓ they do not otherwise; the scale channel is therefore not
194 a finite-sample nuisance but a direct consequence of the working covariance:
195 r departs from one whenever $V_0 \neq V$, and no quantity of data removes the
196 discrepancy.

197 This is also what makes a model ladder informative. Each rung is a different work-
198 ing covariance V_0 fitted to the same data, so tracking r along a graded sequence of
199 working covariances, from the simplest to the production model, isolates the rung,
200 that is the covariance component, at which r departs from one. Because every
201 rung is fitted to the identical simulated data, the comparison is free of Monte
202 Carlo noise between models.

203 2.3 The sample-size gradient

204 A standard-error mis-calibration may be one of two qualitatively different things.
205 It may be a finite-sample artefact, in which case it attenuates as the sample grows
206 and the working-covariance parameters are estimated more sharply; the small-
207 sample corrections of Kenward and Roger^{15,16} address mis-calibration of this kind.
208 Or it may be asymptotic, a consequence of a working covariance that does not
209 match the data-generating covariance no matter how much data is collected, in
210 which case it persists. The two are distinguished by running the decomposition
211 across a range of sample sizes. Attenuation toward nominal calibration marks
212 the finite-sample case; a flat profile marks the asymptotic case. The distinction
213 matters, because a finite-sample artefact at a realistic sample size may be tolerable,
214 whereas an asymptotic bias is a property of the method and will not be outgrown.

215 3 Worked example: an interaction test under carry- 216 over

217 The example is the test of a biomarker-by-treatment interaction in the carryover-
218 sensitivity simulation study of the pmsimstats-ng project. Aggregated N-of-1
219 trials repeatedly cross each participant between an active treatment and a control
220 condition, and the scientific question is whether a baseline biomarker moderates
221 the treatment effect. Carryover, the persistence of drug effect into a washout pe-
222 riod, complicates the analysis, and the study compares analysis-model strategies

for accommodating it¹³. The reporting follows the ADEMP structure of Morris,
White, and Crowther¹³.

3.1 The analysis model

For a symptom score $S_{x,it}$ on participant i at timepoint t , the analysis model is the linear mixed model

$$S_{x,it} = \beta_0 + \beta_1 \text{bm}_i + \beta_2 t + \beta_3 X_{it} + \theta (\text{bm}_i \times X_{it}) + b_i + \varepsilon_{it},$$

fitted by `nlme::lme` with a participant random intercept $b_i \sim N(0, \sigma_b^2)$ and a continuous-time first-order autoregressive residual correlation, `corCAR1(form = ~ t | ptID)`. Here bm_i is the standardised biomarker, X_{it} a drug-exposure predictor, and θ the interaction coefficient of interest. The continuous-time autoregressive residual was adopted in the project for its clinical realism, nearby measurements being more strongly correlated than distant ones, and for the numerical range it affords¹¹. The Wald test of θ is the object of this paper.

3.2 The null configuration

Calibration is a null-hypothesis property, so the diagnosis is run where the biomarker exerts no moderation, $c_{bm} = 0$, and the true interaction is therefore $\theta = 0$. The primary cell is the reference configuration of the parent study: the Hybrid N-of-1 design, $N = 70$ participants, an exponential carryover form with a one-week half-life, and an autoregressive parameter of 0.7. A preliminary check established that at $c_{bm} = 0$ the study's two data-generating architectures collapse to a bit-identical process, so the diagnosis need not stratify by architecture.

The feature of the example that matters for the argument is the relationship between the working covariance and the truth. The simulated symptom score is a sum of three latent factors, each following its own autoregressive process, with timepoint-dependent variances, plus a participant-level baseline term. The working covariance of the analysis model, one random intercept and one single-parameter `corCAR1` residual, cannot reproduce that structure exactly. The example is therefore an instance of the general situation the protocol is designed for: a parametric working covariance that is plausible, conventional, and not exactly correct.

4 Results

The diagnosis was carried out at 5000 null replicates for the four-channel decomposition and the model ladder, and at 1500 replicates per point for the sample-size gradient. The cluster-robust remedy was verified at 3000 replicates. A rejection rate estimated from n_{sim} replicates carries a Monte Carlo standard error of $\sqrt{p(1-p)/n_{\text{sim}}}$ ¹³; at a type-I error near 0.03 this is approximately 0.0024 at 5000 replicates, 0.0031 at 3000, and 0.0044 at 1500, and the comparisons to the nominal level reported below are read against those tolerances. All replicate-level outputs

Table 1: Four-channel decomposition at the primary null cell (Hybrid design, $N = 70$, $c_{bm} = 0$, 5000 replicates), for the three analysis specifications A1, A2, A3. The interaction estimate is effectively unbiased; the SE ratio exceeds one; the statistic is correspondingly compressed; the t and z references coincide because the degrees of freedom are large.

Spec	Mean estimate	SE ratio r	SD(T)	Type I (t)	Type I (z)	Mean DF
A1	+0.032	1.069	0.932	0.033	0.034	487
A2	+0.039	1.097	0.909	0.029	0.029	487
A3	+0.033	1.061	0.939	0.035	0.036	486

are retained for inspection.

4.1 The conservatism is standard-error over-estimation

Table 1 reports the decomposition for the three analysis specifications carried by the parent study, which differ only in the drug-exposure predictor X_{it} and are treated here as three instances of the same interaction test. The reading is consistent across all three. The interaction estimate is effectively unbiased: the null mean falls between +0.03 and +0.04 across the three specifications, on a coefficient whose sampling standard deviation is near unity, a displacement of about 0.035 of a standard deviation. A displacement of that size is statistically detectable at 5000 replicates but is negligible in magnitude, and in any case it points toward rejection and so cannot produce a conservative test. The degrees-of-freedom channel is excluded directly: the model assigns approximately 487 degrees of freedom to the interaction term, and recomputing the p-values against a normal reference moves the realised type-I error by about 0.001, as the near-identical t - and z -reference columns show.

The discrepancy is in the standard-error channel. The model-based standard error exceeds the empirical sampling standard deviation of the estimate by 6 to 10 percent, and the standard deviation of the test statistic agrees with the reciprocal of that ratio to within half a percent, as the identity $SD(T) \approx SD(Z^*)/r$ with $SD(Z^*) = 1$ leads us to expect. With the other three channels inert, $\delta \approx 0$, ν large, and Z^* standard normal, the realised type-I error of Section 2.1 reduces to the closed form

$$\alpha_{\text{real}} \approx 2 \Phi(-r t_{\nu, 1-\alpha/2}),$$

the probability that a standard normal exceeds the threshold $r t_{\nu, 1-\alpha/2}$ rather than $z_{1-\alpha/2}$. Differentiating this closed form exposes the leverage of the scale channel. Near $r = 1$, with ν large enough that $t_{\nu, 1-\alpha/2} \approx z_{1-\alpha/2} = 1.96$,

$$\frac{d \alpha_{\text{real}}}{d r} = -2 t_{\nu, 1-\alpha/2} \phi(r t_{\nu, 1-\alpha/2}) \approx -0.23,$$

so to first order $\alpha_{\text{real}} \approx 0.05 - 0.23(r - 1)$: the realised error falls roughly 2.3 times as fast as the standard error is inflated. A six-to-ten percent over-statement of the

standard error therefore depresses the type-I error into the low 0.03 range, which is the sense in which a numerically modest mis-calibration of the standard error is a material mis-calibration of the test. For the standard-error ratios of Table 1 this predicts a type-I error of 0.031 to 0.037 across the three specifications, matching the observed 0.029 to 0.035 (at $\alpha = 0.05$) to within Monte Carlo error. The scale channel alone, with no contribution from the other three, reproduces the conservatism.

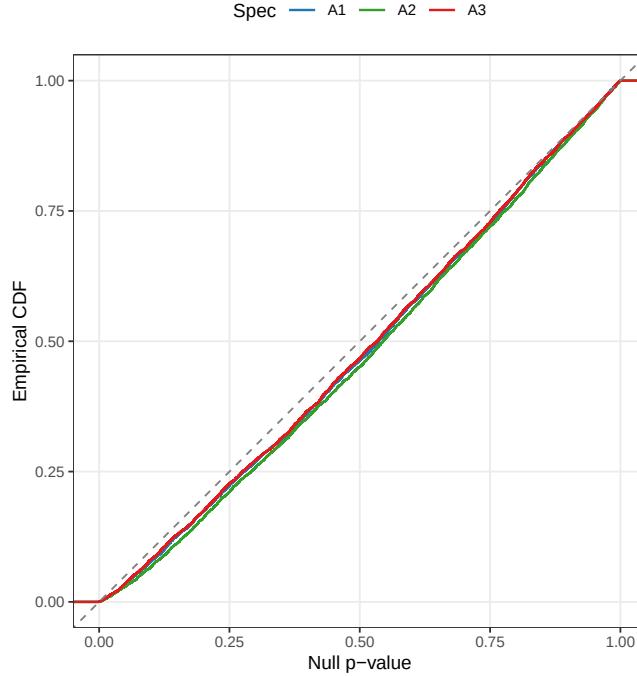


Figure 1: Empirical distribution function of the null p-values at the primary cell, against the Uniform(0,1) diagonal. All three specifications bow uniformly below the diagonal, the signature of a global scale compression rather than a tail anomaly.

Figure 1 confirms the nature of the mis-calibration. The empirical distribution function of the null p-values lies below the diagonal across the whole unit interval and for all three specifications. A test that failed only in its tails would depart from the diagonal only near zero; the uniform bowing seen here is the signature of a global scale compression, which is what an inflated standard error produces. The positive-definiteness correction available to the data generator was not triggered in any of the cell's paths and is excluded as a contributor.

4.2 The corCAR1 residual layer is the cause

The decomposition says the standard error is over-estimated. The model ladder, Table 2, says which part of the model does the over-estimating. Four working covariances were fitted to the identical null datasets: ordinary least squares with no within-cluster term (M0), a random intercept alone (M1), a corCAR1 residual alone (M2), and the production combination of a random intercept with a corCAR1 residual (M3).

Table 3: Sample-size gradient under the production model M3 (1500 replicates per point). The standard-error ratio r and the type-I error (T1) are shown for each specification. Neither trends toward its nominal value as N increases.

N	r A1	r A2	r A3	T1 A1	T1 A2	T1 A3
35	1.072	1.103	1.064	0.038	0.028	0.039
70	1.056	1.083	1.050	0.043	0.031	0.045
140	1.086	1.098	1.079	0.041	0.033	0.042
280	1.052	1.070	1.046	0.037	0.028	0.035

contrast, and ordinary least squares charges it the full residual variance, between-participant and within-participant combined, when the error relevant to a within-participant contrast is only the within-participant part. In the idealised case of a pure within-participant contrast with exchangeable errors the inflation is the variance ratio $r_{\text{OLS}}^2 \approx (\sigma_b^2 + \sigma_\epsilon^2)/\sigma_\epsilon^2$, the total residual variance over its within-participant component, exceeding one by the random-intercept variance σ_b^2 . The familiar warning, the Moulton pitfall, concerns between-cluster effects¹⁷; for a within-cluster effect the sign of the OLS error is reversed. The second feature is that the corCAR1 residual without a random intercept (M2) is intermediate. Taken together the ladder shows that calibration is recovered by the random intercept alone and removed by the corCAR1 addition, and that the magnitude and even the sign of the mis-calibration is a property of the working covariance, not a universal constant.

4.3 The conservatism is asymptotic, not a small-sample artefact

If the conservatism were a finite-sample artefact, it would diminish as the sample grew and the working-covariance parameters were estimated more precisely. Table 3 and Figure 3 show that it does not. Across an eightfold range of sample size, from $N = 35$ to $N = 280$ participants, the standard-error ratio holds between 1.05 and 1.10 with no trend toward one, and the type-I error holds between 0.028 and 0.045 with no trend toward 0.05. At the largest sample size the production specification still rejects the true null at 0.028. The conservatism is asymptotic. It is a property of the M3 working covariance and will not be outgrown.

The explanation that is consistent with all three results is a working-covariance mismatch rather than a small-sample effect. The simulated symptom score is a mixture of autoregressive factors with unequal, timepoint-dependent variances, plus a participant baseline term. The M3 working covariance, a single random intercept and a single-parameter corCAR1 residual, cannot match a mixture of autoregressive processes exactly. Increasing the number of participants estimates the parameters of that working covariance ever more sharply, but it does not change the fact that the covariance being estimated is the wrong one, and so the model-based standard error converges to a biased target. The bias persists because it is a bias of the model, not of the estimate of the model.

form in the empirical cluster cross-products,

$$\hat{V}_{\text{CR}} = (X'X)^{-1} \left(\sum_i X_i' A_i \hat{e}_i \hat{e}_i' A_i X_i \right) (X'X)^{-1},$$

and the family CR0, CR1, CR2, CR3 is indexed by the adjustment matrices A_i ^{9,10}. The unadjusted choice $A_i = I$ is consistent as the number of clusters grows but biased downward at moderate cluster counts; the jackknife choice $A_i = (I - H_{ii})^{-1}$ over-corrects and is, in the marginal setting, the estimator of Mancl and DeRouen⁷. The CR2 choice sets $A_i = (I - H_{ii})^{-1/2}$, the symmetric square root that renders $\hat{V}_{\text{CR}}$ exactly unbiased for the true variance when the working covariance is in fact correct; it is the cluster generalisation of the HC2 estimator of MacKinnon and White²¹. For a mixed model fitted by `lme` the construction is applied in the metric of the working covariance, that is to the generalized-least-squares-transformed design and residuals, following Pustejovsky and Tipton¹⁰, and is recommended with a matching small-sample reference by Imbens and Kolesar²². Because $\hat{V}_{\text{CR}}$ is itself a quadratic form in approximately normal residuals, its null distribution is approximated by a scaled χ^2 , and the Satterthwaite degrees of freedom are recovered by matching its first two moments, $\nu = (\sum_g \lambda_g)^2 / \sum_g \lambda_g^2$ in the eigenvalues λ_g of the matrix defining that form^{10,14}.

Table 4 reports the verification. The model-based standard-error ratio reproduces the conservatism, between 1.05 and 1.10. The CR2 cluster-robust ratio is close to one, between 0.97 and 1.00, and the realised type-I error rises from approximately 0.03 under the model-based test to between 0.045 and 0.049 under the CR2 test, within Monte Carlo error of the nominal 0.05. The Satterthwaite degrees of freedom for the cluster-robust test are approximately 24, far below the model-based 487 and well below the participant count of 70, since they measure the heterogeneity of cluster leverage on the interaction contrast rather than a count of clusters²²; the small-sample reference is doing real work, and a naive normal reference for the sandwich statistic would itself be mildly anti-conservative^{9,23}.

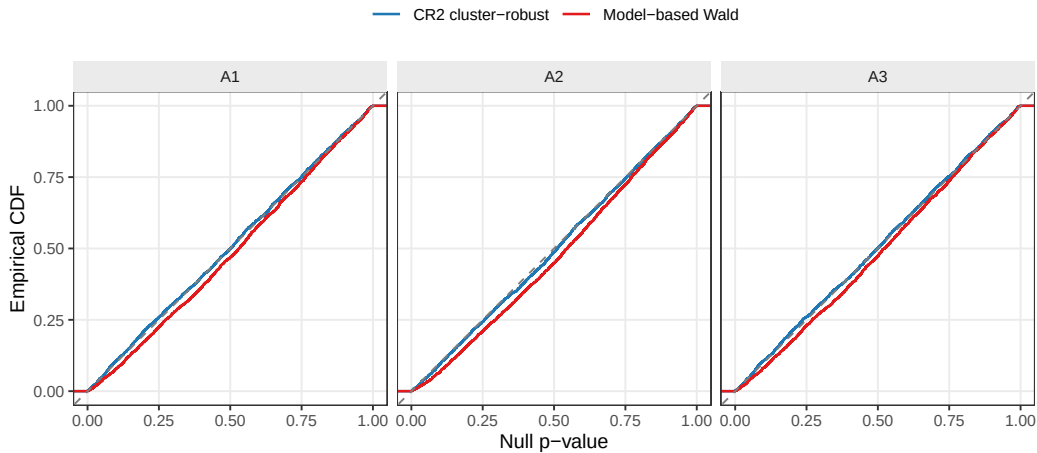


Figure 4: Null p-value distribution functions under the model-based and CR2 cluster-robust tests, by specification. The model-based curves bow below the diagonal; the CR2 curves lie on it.

Table 5: The cluster-robust remedy at an alternative cell (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, 3000 replicates). Power is the rejection rate of the interaction test; the standard-error ratio r is as defined for the null. The model-based test is conservative here too, and the CR2 test recalibrates it.

Spec	r model-based	r CR2	Power model	Power CR2
A1	1.124	0.986	0.757	0.795
A2	1.152	0.988	0.843	0.878
A3	1.119	0.989	0.755	0.788

Figure 4 shows the same result as a distribution function. The model-based p-values bow below the diagonal, as in Figure 1; the cluster-robust p-values lie on it. The point estimate and the corCAR1 model are unchanged; only the standard error of the interaction, and the reference against which it is tested, are replaced.

4.5 The remedy at an alternative cell

The diagnosis to this point has been conducted under the null, where calibration is defined. A reader of a comparative study, however, will press a further question. The purpose of the parent study is to rank analysis specifications by power, and if a recalibration of the standard error raises the power of every specification, does it also disturb their order? We checked this directly at one alternative cell, the parent study's reference configuration with a non-null biomarker moderation, $c_{bm} = 0.45$, retaining the Architecture-B Hybrid design at $N = 70$.

Three things follow from Table 5. First, the standard-error over-estimation is not confined to the null: at the alternative cell the model-based ratio is 1.12 to 1.15, if anything slightly larger than the 1.06 to 1.10 seen under the null. Second, the CR2 cluster-robust standard error recalibrates the alternative cell as it did the null, bringing the ratio to 0.99 for all three specifications. Third, and most consequential for a comparative study, the recalibration is close to a uniform level shift. It raises the power of every specification by roughly three to four percentage points, and the gaps between specifications are left intact: the advantage of the leading specification A2 over A1 is +0.086 under the model-based test and +0.082 under the cluster-robust test, and over A3 it is +0.088 and +0.089. The ranking, and the magnitude of the leading specification's advantage, survive the recalibration unchanged.

The lesson generalises beyond the example. A cluster-robust standard error corrects the absolute error rate, the type-I error under the null and the power under the alternative alike, but because the correction acts on the standard error and not on the point estimate, and because the standard-error mis-calibration here is similar across the specifications being compared, it shifts the level without re-ordering the contrast. A comparative study whose claim is a ranking is therefore robust to the recalibration, even where its absolute power figures, as reported

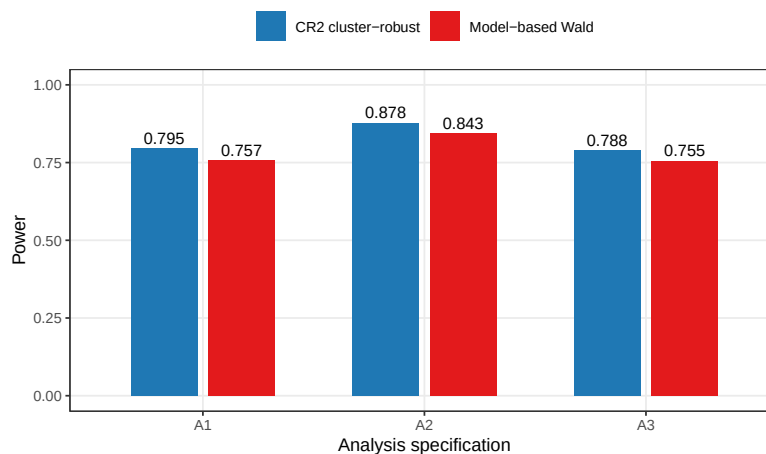


Figure 5: Power of the three analysis specifications at the alternative cell, under the model-based and CR2 cluster-robust tests. The recalibration raises every specification by a near-uniform increment and preserves their order.

under the model-based test, are mildly pessimistic.

5 Discussion

5.1 The result is not about autoregressive correlation

It would be a mistake to read this paper as a caution against the continuous-time autoregressive residual in particular. The `corCAR1` term is the component the model ladder happened to implicate in this example, but the mechanism is general. The model ladder is itself the evidence: it fitted four working covariances and obtained four different standard-error ratios, from 1.3 under ordinary least squares to 1.0 under a random intercept to near 1.1 under the production model. The mis-calibration is a property of the match between the working covariance and the truth, and any parametric residual or random-effects structure, compound symmetry, an unstructured form, a random slope, a spatial correlation, is capable of producing it. That a mis-specified working covariance leaves the model-based fixed-effect standard error inconsistent is, moreover, not a new claim; it is the Eicker-Huber-White result carried into the mixed-model setting^{3-5,24}. What the worked example contributes is a quantified, decomposed instance in a realistic longitudinal interaction test, with two features that are not obvious in advance.

5.2 Two features that are not obvious in advance

The first is the direction. The sandwich literature is usually motivated by anti-conservative tests: ignored positive correlation under-states standard errors, and the robust estimator corrects an inflated rejection rate downward. Here the model-based test is conservative, and the cluster-robust estimator corrects the rejection rate upward. The ladder explains why the intuition can mislead. The effect under test is a within-cluster contrast, and for a within-cluster contrast the relationship between a mis-specified covariance and the standard-error error need not have

the sign that the between-cluster intuition supplies. An analyst who reasons that ‘my standard errors are conservative because I have modelled the correlation’ has the logic backward in this case: modelling the correlation, through the `corCAR1` term, is what introduced the conservatism, and the less elaborate model was the calibrated one.

The second is that the bias is asymptotic. It would have been comfortable to attribute the conservatism to moderate sample size, and an earlier informal reading of the parent study did so. The sample-size gradient refutes that reading directly. This matters for how the problem is classified. The small-sample corrections of Kenward and Roger^{15,16} adjust the fixed-effect standard error and its degrees of freedom for the finite-sample uncertainty in the estimated variance parameters, operationalising the variance-underestimation result of Kackar and Harville²⁵; they are the right instrument for a finite-sample artefact. They are not the right instrument here, because the bias is not a finite-sample artefact: it does not vanish as the variance parameters are estimated with unlimited precision. A Kenward-Roger correction to a mis-specified working covariance corrects the finite-sample term and leaves the asymptotic mismatch untouched. The cluster-robust standard error addresses the asymptotic mismatch directly, which is why it is the appropriate remedy here.

5.3 When the problem should be expected

The conditions for the problem are not exotic. They are a fixed-effect interaction tested in a mixed model; a parametric working covariance that is plausible but not known to be exactly correct, which is to say almost every parametric working covariance in applied use; and an interaction that is partly or wholly a within-cluster contrast. Wherever those conditions hold, the calibration of the model-based Wald test is an open question, and the staged protocol of Section 2 answers it at the cost of a simulation. We would suggest, as a matter of routine practice, that an interaction test reported from a mixed model with a non-trivial working covariance be accompanied either by a calibration check of this kind or by a cluster-robust standard error, which removes the question by being agnostic to the working covariance. This accords with a broader case for giving robust and marginal methods more routine consideration alongside the conditional mixed model²⁶.

5.4 The marginal-model alternative

The remedy adopted above keeps the conditional mixed model and adds a cluster-robust standard error after the fact. A reader may reasonably ask whether the marginal route, generalized estimating equations, would not be more natural, since the method carries the sandwich variance from the outset rather than bolting it on⁶. It would. For a linear model with an identity link the marginal and conditional fixed-effect coefficients coincide, so a generalized estimating equations analysis estimates the same interaction and answers the same question; the working correlation, exchangeable or autoregressive, governs only the efficiency of the

point estimate and not the validity of the inference.

We checked the marginal route at the primary null cell, fitting the generalized estimating equations counterpart of the analysis model with an AR(1) working correlation. The naive sandwich standard error proved mildly anti-conservative, the small-sample behaviour the literature describes for a moderate number of clusters: across the three specifications it under-estimated the sampling standard deviation by roughly four percent, and the realised type-I error was 0.067 to 0.069. A bias-corrected sandwich of the kind introduced by Mancl and DeRouen⁷ removed most of the discrepancy, bringing the standard-error ratio to within one percent of one and the type-I error to 0.055 to 0.058. The marginal route is therefore viable, but it carries the same lesson as the conditional one: at the cluster counts typical of aggregated N-of-1 work the sandwich variance itself needs a small-sample correction. The two corrected tests err in opposite and modest directions, the conditional CR2 test slightly conservative and the marginal Mancl-DeRouen test slightly anti-conservative, and either is a defensible calibrated analysis. The choice between them is practical rather than principled: the conditional route preserves an existing mixed-model pipeline and changes only the standard error, while the marginal route is conceptually cleaner but a more thoroughgoing change of model.

5.5 Limitations

Several limitations bound the scope of the evidence. The diagnosis was carried out for one data-generating family, one estimand, and one production working covariance; the asymptotic conservatism is demonstrated for the corCAR1-plus-random-intercept model, and the generality beyond that rests on the model ladder and on established theory rather than on a sweep of every covariance structure. A broader empirical base, fitting an unstructured or compound-symmetry analysis to the same data, would strengthen the general claim, and is a natural extension. The cluster-robust remedy is not without cost: the Satterthwaite degrees of freedom for the CR2 test are approximately 24 against a model-based 487, so the robust test spends precision to buy validity, and at a genuinely small number of clusters that price would rise. The principal diagnosis was conducted at the null, where calibration is defined; the consequence for power was checked at a single alternative cell (Section 4.5), where the recalibration proved a near-uniform level shift that preserved the ranking of specifications. Whether that uniformity holds across the whole factorial, rather than at the one cell examined, was not established. Finally, the small positive point-estimate bias noted in Section 4.1, while negligible in magnitude and irrelevant to the conservatism, is a genuine finite-sample feature of the estimator that a separate examination might characterise.

6 Conclusions

1. The model-based Wald test of a fixed-effect interaction in a linear mixed model can be materially mis-calibrated when the parametric working covari-

ance does not match the data-generating covariance. In the worked example the mis-calibration is a conservative one: the model-based standard error of the interaction is over-estimated by 6 to 10 percent and the realised type-I error is approximately 0.03 against a nominal 0.05.

2. The four-channel decomposition localises the discrepancy to the standard-error channel and excludes point-estimate bias, the degrees-of-freedom rule, and the positive-definiteness correction.
3. The working-covariance model ladder localises the cause to the corCAR1 residual layer. A random intercept alone is well calibrated; adding the corCAR1 residual removes the calibration.
4. The sample-size gradient shows the mis-calibration is asymptotic. It does not attenuate from $N = 35$ to $N = 280$, and so it is a property of the working covariance and not a small-sample artefact amenable to a Kenward-Roger correction.
5. A CR2 bias-reduced cluster-robust standard error, clustered on participant and referred to a Satterthwaite distribution, restores the standard-error ratio to approximately one and the type-I error to approximately 0.05, without altering the point estimate or the analysis model. At an alternative cell the same recalibration is a near-uniform level shift in power that preserves the comparative ranking of specifications.
6. The staged protocol, the four-channel decomposition, the model ladder, and the sample-size gradient, is a reusable instrument for any mixed-model interaction test whose calibration is in doubt. We recommend that interaction tests reported from mixed models with a non-trivial working covariance be accompanied by a calibration check of this kind, or by a cluster-robust standard error that is agnostic to the working covariance.

7 Reproducibility

The diagnosis is driven by six scripts under `analysis/scripts/carryover-sensitivity/`: `diagnose-null-type1.R` for the four-channel decomposition, `diagnose-null-type1-stages46.R` for the model ladder and the sample-size gradient, `diagnose-null-type1-cr2.R` for the cluster-robust verification at the null, `diagnose-alt-cell-cr2.R` for the alternative-cell check, and `diagnose-null-type1-gee.R` for the GEE marginal-model check, with `check-null-architecture-equivalence.R` establishing the architecture-invariance of the null. Each writes a replicate-level `.rds` file into the `output/` subdirectory, and this manuscript reads those files and regenerates every table and figure inline. The cluster-robust standard errors use the `clubSandwich` package. The examination plan that preceded the manuscript is retained alongside it as `type1-examination-plan.md`.

8 References

1. Laird NM, Ware JH. Random-effects models for longitudinal data. *Biometrics*. 1982;38(4):963-974. doi:10.2307/2529876

- 575 2. Diggle PJ, Heagerty P, Liang KY, Zeger SL. *Analysis of Longitudinal Data*.
2nd ed. Oxford University Press; 2002.
- 576 3. Eicker F. Limit theorems for regressions with unequal and dependent errors.
In: *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics
and Probability, Volume 1*. University of California Press; 1967:59-82.
- 577 4. Huber PJ. The behavior of maximum likelihood estimates under nonstan-
dard conditions. In: *Proceedings of the Fifth Berkeley Symposium on
Mathematical Statistics and Probability, Volume 1*. University of Califor-
nia Press; 1967:221-233.
- 578 5. White H. A heteroskedasticity-consistent covariance matrix estimator and
a direct test for heteroskedasticity. *Econometrica*. 1980;48(4):817-838.
doi:10.2307/1912934
- 579 6. Liang KY, Zeger SL. Longitudinal data analysis using generalized linear
models. *Biometrika*. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13
- 580 7. Mancl LA, DeRouen TA. A covariance estimator for GEE with
improved small-sample properties. *Biometrics*. 2001;57(1):126-134.
doi:10.1111/j.0006-341X.2001.00126.x
- 581 8. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regres-
sion with multi-stage samples. *Survey Methodology*. 2002;28(2):169-181.
- 582 9. Cameron AC, Miller DL. A practitioner's guide to cluster-robust inference.
Journal of Human Resources. 2015;50(2):317-372. doi:10.3368/jhr.50.2.317
- 583 10. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust
variance estimation and hypothesis testing in fixed effects mod-
els. *Journal of Business & Economic Statistics*. 2018;36(4):672-683.
doi:10.1080/07350015.2016.1247004
- 584 11. Pinheiro JC, Bates DM. *Mixed-Effects Models in S and S-PLUS*. Springer;
2000.
- 585 12. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing ag-
gregated N-of-1 trial designs for predictive biomarker validation: Statis-
tical methods and practical considerations. *Frontiers in Digital Health*.
2020;2:13. doi:10.3389/fdgth.2020.00013
- 586 13. Morris TP, White IR, Crowther MJ. Using simulation studies to evalu-
ate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102.
doi:10.1002/sim.8086

- 587 14. Satterthwaite FE. An approximate distribution of estimates of variance components. *Biometrics Bulletin*. 1946;2(6):110-114. doi:10.2307/3002019
- 588 15. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*. 1997;53(3):983-997. doi:10.2307/2533558
- 589 16. Kenward MG, Roger JH. An improved approximation to the precision of fixed effects from restricted maximum likelihood. *Computational Statistics & Data Analysis*. 2009;53(7):2583-2595. doi:10.1016/j.csda.2008.12.013
- 590 17. Moulton BR. An illustration of a pitfall in estimating the effects of aggregate variables on micro units. *The Review of Economics and Statistics*. 1990;72(2):334-338. doi:10.2307/2109724
- 591 18. Royall RM. Model robust confidence intervals using maximum likelihood estimators. *International Statistical Review*. 1986;54(2):221-226. doi:10.2307/1403146
- 592 19. Tipton E. Small sample adjustments for robust variance estimation with meta-regression. *Psychological Methods*. 2015;20(3):375-393. doi:10.1037/met0000011
- 593 20. Tipton E, Pustejovsky JE. Small-sample adjustments for tests of moderators and model fit using robust variance estimation in meta-regression. *Journal of Educational and Behavioral Statistics*. 2015;40(6):604-634. doi:10.3102/1076998615606099
- 594 21. MacKinnon JG, White H. Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*. 1985;29(3):305-325. doi:10.1016/0304-4076(85)90158-7
- 595 22. Imbens GW, Kolesár M. Robust standard errors in small samples: Some practical advice. *The Review of Economics and Statistics*. 2016;98(4):701-712. doi:10.1162/REST_a_00552
- 596 23. Long JS, Ervin LH. Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*. 2000;54(3):217-224. doi:10.1080/00031305.2000.10474549
- 597 24. Freedman DA. On the so-called Huber sandwich estimator and robust standard errors. *The American Statistician*. 2006;60(4):299-302. doi:10.1198/000313006X152207

- 598 25. Kackar RN, Harville DA. Approximations for standard errors of estimators of fixed and random effects in mixed linear models. *Journal of the American Statistical Association*. 1984;79(388):853-862. doi:10.1080/01621459.1984.10477102
- 599 26. McNeish D, Stapleton LM, Silverman RD. On the unnecessary ubiquity of hierarchical linear modeling. *Psychological Methods*. 2017;22(1):114-140. doi:10.1037/met0000078

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